

Transparent 360-Degree Display for High-Resolution Naked-Eye Stereoscopic Aerial Images

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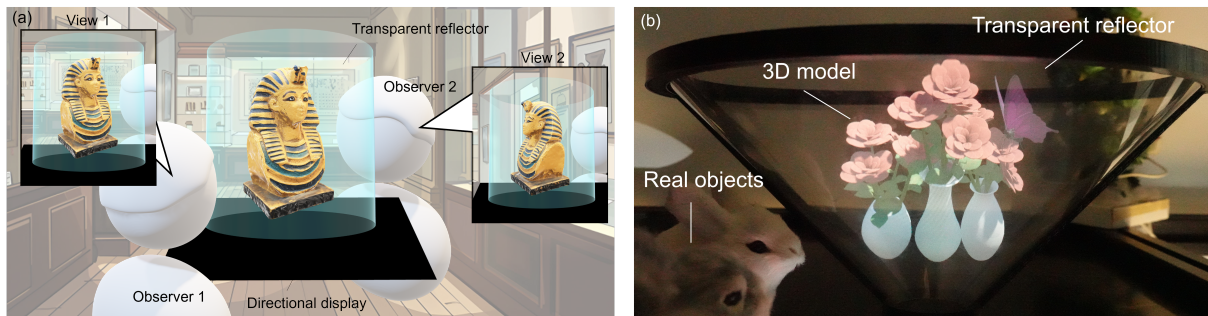


Figure 1: Overview of the mid-air imaging method: (a) Image of a 360-degree naked-eye stereoscopic image formed in mid-air. (b) Image formed in mid-air from the observer's viewpoint ("single-eye view").

Abstract

We developed a display that presents high-resolution stereoscopic images floating in mid-air. Conventional methods have not yet achieved the presentation of realistic images floating in mid-air due to issues such as resolution and the prominence of the display medium. Our method realizes a thin directional display that can present directional images at the observer's position and display high-resolution naked-eye stereoscopic images. Additionally, by combining our method with a curved transparent reflective panel, we can present stereoscopic images in mid-air that are observable from all directions.

CCS Concepts

• Computing methodologies → Mixed / augmented reality.

Keywords

Auto stereoscopy, Transparent Display, Stereoscopic Display, 3D

ACM Reference Format:

Mari Shiina and Naoki Hashimoto. 2024. Transparent 360-Degree Display for High-Resolution Naked-Eye Stereoscopic Aerial Images. In *SIGGRAPH Asia 2024 Posters (SA Posters '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681756.3697887>

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SA Posters '24, December 03–06, 2024, Tokyo, Japan

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ACM ISBN 979-8-4007-1138-1/24/12

<https://doi.org/10.1145/3681756.3697887>

1 INTRODUCTION

For a long time, there has been a desire to realize scenes often depicted in science fiction movies where realistic images of people and objects float in mid-air. As attempts to achieve such aerial stereoscopic images, stereoscopic and aerial image technologies have been developed. However, stereoscopic technologies, including light field displays, do not consider blending image projections to float in the air. The images are only realized in front of the display, and the issue of resolution degradation has not been resolved.

Therefore, methods using Pepper's Ghost technique, an illusion technique for presenting virtual images [Salvador 2019], and methods for forming real images in mid-air [Zhang et al. 2021] have been developed to realize aerial images. However, these methods still have poor resolution.

2 PROPOSED METHOD

2.1 Overall Overview

In this study, we combined a high-resolution directional display with a transparent reflector that has minimal presence to achieve the aerial presentation of highly realistic stereoscopic images. The directional display uses lenses on the light source to present images directionally at specific positions [Figure 2(b)]. By rapidly switching between the light and parallax images with strong directionality for the left and right eyes, unlike conventional methods that present multi-view images simultaneously, we achieve high-resolution, naked-eye stereoscopic images. Furthermore, this directional display is combined with a curved transparent reflector. Using the reflection of the image light from the directional display allows stereoscopic images to be presented within the reflector. Our system tracks the positions of observers around the image and

controls the direction of the reflected light, allowing observers to always seamlessly view the stereoscopic images from all directions.

2.2 Structure of the Directional Display

The structure of the proposed directional display is shown in Figure 2(b). The display consists of a light source, which is the backlight of a general liquid crystal display, and an image presentation unit composed of liquid crystal elements that display images. Additionally, it includes optical elements that change the images according to the viewing position. To achieve directional image presentation, an optical element with an array of plano-convex lenses is placed on the light source. Using these optical elements gives the light source and parallax image presentation directionality.

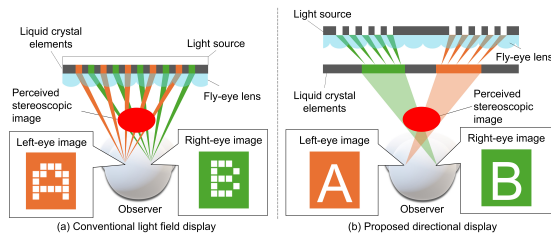


Figure 2: Comparison of stereoscopic presentation methods

By rapidly switching parallax images with directionality for the left and right eyes using time-division multiplexing, we achieve naked-eye stereoscopic images [Figure 2(b)]. In conventional light field displays, optical elements are arranged on liquid crystal elements, and as the number of viewpoints for the stereoscopic image increases, the resolution of the parallax images displayed on the image presentation unit decreases, reducing the resolution of the stereoscopic image [Figure 2(a)]. In our method, by placing optical elements on the light source, we present stereoscopic images with a resolution equivalent to that of the image presentation unit.

2.3 Making the Directional Display Transparent

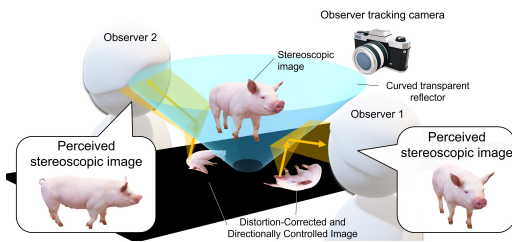


Figure 3: Aerial stereoscopic image presentation

By placing a curved transparent reflector on the directional display and making the display transparent, we achieve aerial image presentation. The curved shape of the reflector also expands the observation range of the stereoscopic images to 360° (Figure 3).

The image light displayed by the directional display is reflected by the reflector and reaches the observer's eyes. With the Pepper's

Ghost technique, the observer perceives the stereoscopic image as if it were within the reflector. To address distortions and directional changes of the stereoscopic image caused by the curved shape of the reflector, we perform distortion correction and directional control of the image using ray tracing.

Furthermore, by tracking the observer, we enable stereoscopic image observation from all positions, including above, below, left, and right. Additionally, it is possible to simultaneously present stereoscopic images to multiple observers.

3 RESULTS

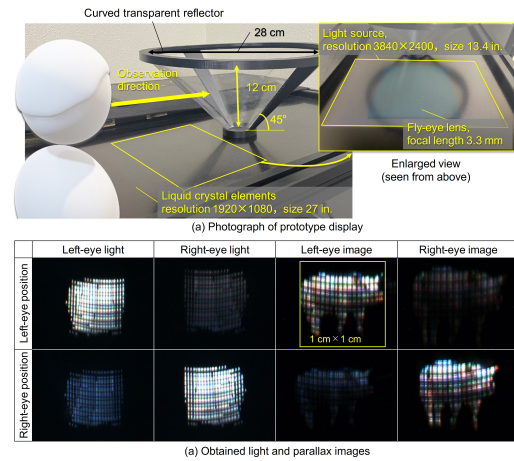


Figure 4: Experimental setup and results

Figure 4 shows the prototype we created. The light source was a 13.4-inch monitor with a resolution of 3840×2400, and a fly-eye lens with a focal length of 3.3 mm was placed on the light source. The image presentation used the liquid crystal elements from a 27-inch monitor with a resolution of 1920×1080, with the backlight removed. An environment similar to that of this prototype was created within the game engine Unity, and distortion correction and directional control of the image were performed using ray tracing. The prototype was tested to verify whether appropriate directionality could be achieved. A camera was placed at the observer's position to capture the light source and parallax images, with its position adjusted to correct for errors in the prototype's placement (Figure 4).

The results confirmed that the directionality for the left and right eyes of the observer was appropriate. However, black lines were observed in the image, indicating a decrease in image quality. This issue is believed to be caused by misalignment between the lens and the display, and calibration of the placement is planned for future improvement.

References

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